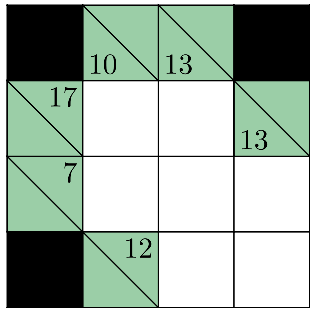
Puzzles and Linear Systems

1. A Kakuro(-ish) Puzzle. Enter numbers from 1 to 9 in any order into the blank cells. The rows and columns should sum to the number on the grid. Within a given row or column, no number can be repeated.



This puzzle is taken from “Linear and Complex Analysis for Applications” by John D.Angelo.

Definition 1. *A* **linear equation** *in the variables* $x_1, \ldots, x_n$ *is an equation that can be put in the following form*

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where b *and the coefficients* $a_1, \ldots, a_n$ *are real or complex numbers. A* **system of linear equations** *is a collection of one or more linear equations involving the same variables. A* **solution** *to the linear system is a list of* $(s_1, \ldots, s_n)$ *of numbers that are solutions to each of the linear equations in the system.*

Given a linear system, some fundamental questions are:

- Does there exist at least one solution?
- If a solution exists, how can we find it?
- If a solution exists, is it unique?

Definition 2. *A linear system is* **consistent** *if there is a solution and* **inconsistent** *if there is no solution.*

2. Find examples of the following.

(a) An inconsistent linear system in two variables with two equations.

(b) A consistent linear system in two variables with two equations.

Fact: A linear system has either no solution, exactly one solution, or infinitely many solutions.

Definition 3. *The following elementary row operations preserve solutions.*

(a) Replace one row by the sum of itself and a multiple of another row.

(b) Interchange two rows.

(c) Multiply all entries in a row by a non-zero constant.

Definition 4. *Two matrices are called* **row equivalent** *if there is a sequence of elementary row operations changing one matrix to another.*

Fact: If the augmented matrices of two linear systems are row equivalent, then they both have the same solution set.

3. Solve the following systems.

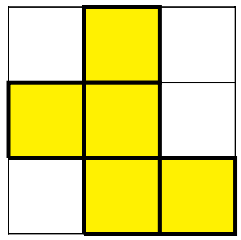
(a)

$$\begin{aligned} 2x + 3y &= 0 \\ 4x + 5y &= 1 \end{aligned}$$

(b)

$$\begin{aligned} 2x + 3y &= 0 \\ 4x + 6y &= 1 \end{aligned}$$

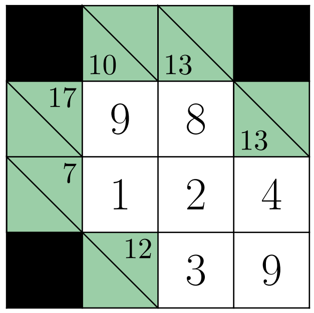
4. Lights Out Puzzle. The yellow squares represent rooms with the lights on, and the white squares represent squares with lights off. The goal is to turn off all the lights, but there is a catch. When you flip the switch in any room, the lights in the rooms horizontally and vertically adjacent are also flipped.

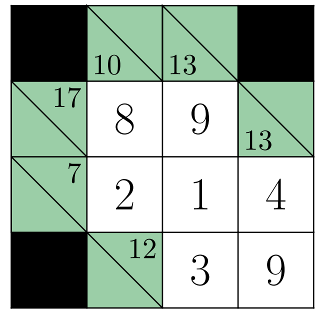


5. Write a short summary of today's key ideas.

Puzzles and Systems – Answers / Solutions

1. Here are the two possible solutions:





This puzzle is taken from "Linear and Complex Analysis for Applications" by John D'Angelo.

2.

(a) An inconsistent linear system in two variables with two equations.

$$\begin{aligned} y - x &= 0 \\ y - x &= 1 \end{aligned}$$

(b) A consistent linear system in two variables with two equations with infinitely many solutions.

$$\begin{aligned} y - x &= 0 \\ 2y - 2x &= 0 \end{aligned}$$

(c) A consistent linear system in two variables with two equations with exactly one solution.

$$\begin{aligned} y - x &= 0 \\ y + x &= 0 \end{aligned}$$

3.

(a) We begin by translating the linear system into the corresponding augmented matrix:

$$\left(\begin{array}{cc|c} 2 & 3 & 0 \\ 4 & 5 & 1 \end{array}\right)$$

Now we can perform elementary row operations to simplify the matrix:

$$\left(\begin{array}{cc|c} 2 & 3 & 0 \\ 4 & 5 & 1 \end{array}\right) \xrightarrow{R_2 - 2R_1} \left(\begin{array}{cc|c} 2 & 3 & 0 \\ 0 & -1 & 1 \end{array}\right) \xrightarrow{-R_2} \left(\begin{array}{cc|c} 2 & 3 & 0 \\ 0 & 1 & -1 \end{array}\right) \xrightarrow{R_1 - 3R_2} \left(\begin{array}{cc|c} 2 & 0 & 3 \\ 0 & 1 & -1 \end{array}\right) \xrightarrow{\frac{1}{2}R_1} \left(\begin{array}{cc|c} 1 & 0 & \frac{3}{2} \\ 0 & 1 & -1 \end{array}\right),$$

which becomes the following system of equations:

$$\begin{aligned} x &= \frac{3}{2} \\ y &= -1 \end{aligned}$$

(b) We begin again by translating the linear system into the corresponding augmented matrix:

$$\left(\begin{array}{cc|c} 2 & 3 & 0 \\ 4 & 6 & 1 \end{array}\right)$$

Now we can perform elementary row operations to simplify the matrix:

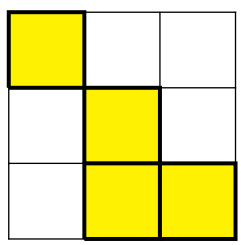
$$\left(\begin{array}{cc|c} 2 & 3 & 0 \\ 4 & 6 & 1 \end{array}\right) \xrightarrow{R_2-2R_1} \left(\begin{array}{cc|c} 2 & 3 & 0 \\ 0 & 0 & 1 \end{array}\right),$$

which becomes the following system of equations:

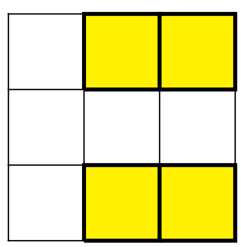
$$\begin{aligned} 2x + 3y &= 0 \\ 0 &= 1. \end{aligned}$$

Thus the system has no solutions and it is inconsistent. Geometrically, we can see that the equations are parallel lines.

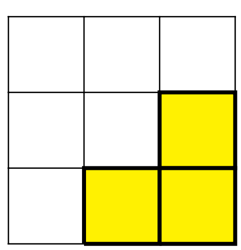
4. We start in the upper-left corner:



Next we turn the upper middle light to get the following:



Now we turn the light in the upper right to get the following:



Now going in the lower right turns off all the lights.